

GEOMETRIC DIMENSIONING AND TOLERANCES

UNIT 1 :- BASIC CONCEPTS

TOPICS:- General terms and definitions of geometrical features – General Principle of sizes – System of limits and fits – Inspection of dimensional and geometrical deviations – Datums, Datums systems, and selections of datums. Restraining degrees of freedom, DOF, Simulators. Rule #1 (Boundary principle) and Rule #2.

GENERAL TERMS AND DEFINITIONS OF GEOMETRICAL FEATURES

WHAT IS GD&T?

GD&T, short for Geometric Dimensioning and Tolerancing, is a system for defining and communicating design intent and engineering tolerances that helps engineers and manufacturers optimally control variations in manufacturing processes.

- **General terms and definitions related to geometrical features:-**

Point:

Definition: A location in space with no size or shape.

Line:

Definition: A straight path of points extending infinitely in both directions.

Line Segment:

Definition: A part of a line with two endpoints.

Ray:

Definition: A part of a line that has one endpoint and extends infinitely in one direction.

Plane:

Definition: A flat, two-dimensional surface that extends infinitely in all directions.

Angle:

Definition: The figure formed by two rays with a common endpoint.

Polygon: Definition: A closed geometric figure with straight sides (e.g., triangles, rectangles, pentagons).

Circle:

Definition: The set of all points in a plane that are equidistant from a fixed center.

Arc:

Definition: A segment of a curve, especially a portion of a circle.

Polyhedron:

Definition: A three-dimensional geometric figure with flat faces and straight edges.

Sphere:

Definition: A three-dimensional object where all points on its surface are equidistant from its center.

Cylinder:

Definition: A three-dimensional object with two parallel and congruent circular bases connected by a curved surface.

Cone:

Definition: A three-dimensional object with a circular base and a curved surface that tapers to a point (apex).

Volume:

Definition: The amount of space occupied by a three-dimensional object.

Surface Area:

Definition: The total area of the outer surface of a three-dimensional object.

Diagonal:

Definition: A line segment connecting two non-adjacent vertices of a polygon or polyhedron.

Perimeter:

Definition: The total length of the boundary of a two-dimensional figure.

Center of Mass:

Definition: The point in a body or system of bodies where the mass is concentrated; the balance point.

Symmetry:

Definition: A balanced and proportionate arrangement of parts; a property of a geometric figure that remains unchanged under certain transformations.

Congruent:

Definition: Having the same size and shape.

Geometric Category	Tolerance Type	Description	Symbol	ASME Section
For Individual Features	Form	Straightness		6.4.1
		Flatness		6.4.2
		Circularity		6.4.3
Individual or Related Features	Profile	Line Profile		6.5.2(b)
		Surface Profile		6.5.2(a)
For Related Features	Orientation	Angularity		6.6.2
		Perpendicularity		6.6.4
		Parallelism		6.6.3
	Location	Position		5.2
		Concentricity		5.11.3
		Symmetry		5.13
	Runout	Circular Runout *		6.7.1.2.1
		Total Runout *		6.7.1.2.2

* Arrows may be filled or not filled

General Principle of sizes:-

Basic dimensions are theoretically exact numerical values used to define the form, size, orientation, or location of a part or feature. Basic dimensions are usually shown on a drawing enclosed in a box, but they can also be invoked by referencing a standard or by a note on the drawing.

Basic Dimensions :-Basic dimensions are exact numerical values used to define the theoretically exact size, location, or orientation of a feature. Basic dimensions are denoted by enclosing the dimension value in a rectangle (↻) rather than a circle (⊙), which is used for reference dimensions.

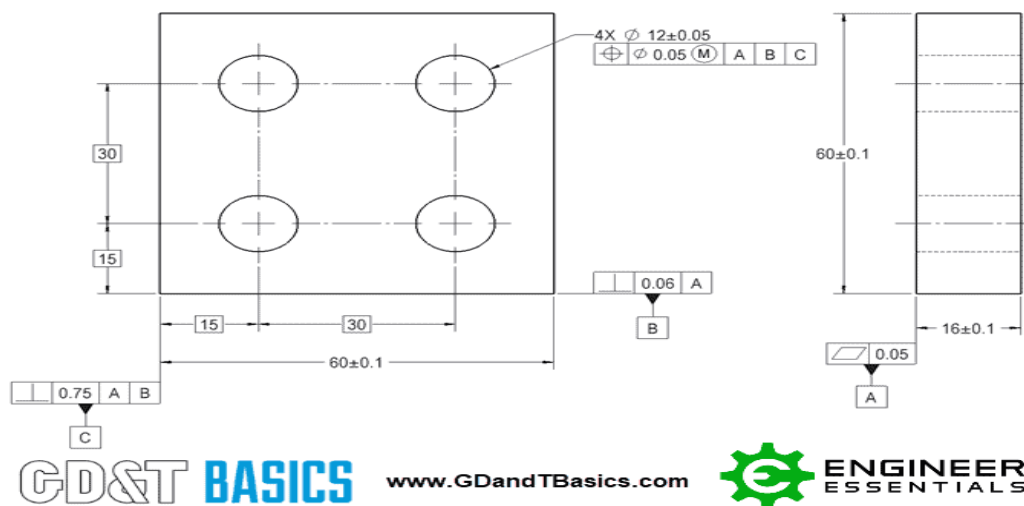
Reference Dimensions: Reference dimensions are used for information purposes and do not govern production or inspection. They are typically enclosed in a circle (ϕ) on the drawing.

Maximum Material Condition (MMC): MMC is a condition where the feature contains the maximum amount of material, either due to the largest hole size or the smallest shaft size. Tolerances are applied to ensure that the part will always fit within its specified limits, regardless of variation.

Least Material Condition (LMC): LMC is a condition where the feature contains the least amount of material, either due to the smallest hole size or the largest shaft size. Tolerances are applied to ensure that the part will always fit within its specified limits, regardless of variation.

Regardless of Feature Size (RFS): RFS is a concept where the tolerance applies at any feature size within the specified limits. It is indicated by the use of the " Δ " symbol in the feature control frame.

Material Boundary: The material boundary is the boundary of a feature at its maximum material condition (MMC). It is used to define the limit within which the feature must be contained.



System of limits and fits:-

The extreme permissible values of a dimension are known as limits. The degree of tightness or looseness between two mating parts that are intended to act together is known as the fit of the parts.

The system of limits and fits in GD&T (Geometric Dimensioning and Tolerancing) is a way of defining the allowable variations in size and geometry for mating parts. While GD&T

primarily focuses on the form, orientation, and location of features, the system of limits and fits complements GD&T by addressing size-related variations.

Basic Size (Nominal Size):

As in traditional limits and fits, the basic size is the nominal or target size for a feature, and GD&T is often applied to ensure that the form, orientation, and location are within acceptable limits.

Tolerance in GD&T:

In GD&T, tolerance is expressed through geometric characteristics and is not only limited to size but also includes form, orientation, and location. Tolerance in GD&T is specified using symbols such as $\perp$ (perpendicularity), $\bigcirc$ (circularity), and $\angle$ (angularity), among others.

Datum Reference Frames (DRF) in GD&T:

GD&T utilizes datum reference frames to establish a coordinate system for defining the geometric characteristics of a part. Datum features and their related datums play a crucial role in controlling the size, orientation, and location of features.

Virtual Condition in GD&T:

Virtual condition is a concept in GD&T that considers the worst-case scenario for the size, form, or orientation of a feature. It is used to establish a relationship between features and is related to both size and geometric tolerancing.

Envelope Requirement in GD&T:

The envelope requirement in GD&T specifies that the entire feature must fit within the specified limits of size and form. This concept ensures that the part will function as intended even with variations in size and form.

Positional Tolerance in GD&T:

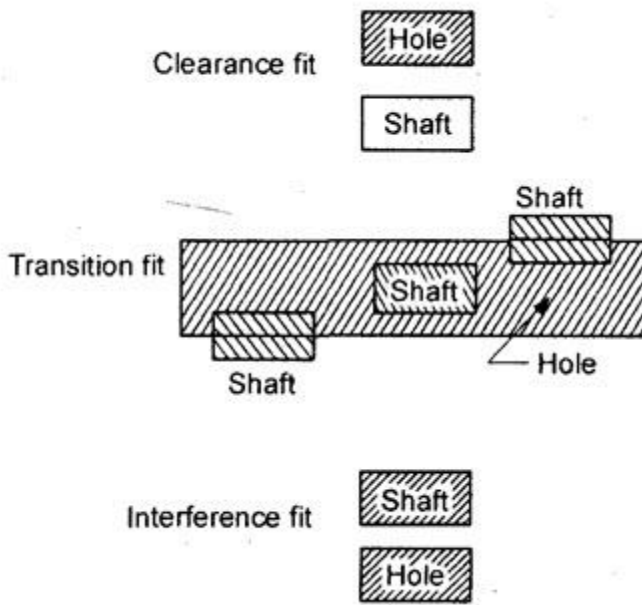
Positional tolerance in GD&T controls the location of features relative to a specified reference point or points. It ensures that the features are within the acceptable limits of size and form while also maintaining proper orientation and location.

Profile Tolerance in GD&T:

Profile tolerance in GD&T controls the shape of a feature within a specified boundary. It considers both size and form, ensuring that the feature conforms to the specified geometric shape.

Concentricity and Symmetry in GD&T:

GD&T uses symbols like $\oplus$ (concentricity) and $\otimes$ (symmetry) to control the distribution of material around a central axis. These symbols help ensure proper alignment and symmetry of features while considering size-related variations.



FITS:-

When two parts are to be assembled the relation resulting from the difference between their sizes before assembly is called a fit. The fit signifies the range of tightness or looseness which may result from the application of a specific combination of allowances and tolerances in the design of mating parts.

Types of Fits:- The three types of fits

- 1. Clearance fit:** A clearance fit is one having limits of size so prescribed that a clearance always results when mating parts are assembled.
- 2. Interference fit:** An interference fit is one having limits of size so prescribed that an interference always results when mating parts are assembled.
- 3. Transition fit:** A transition fit is one having limits of size so prescribed that either a clearance or interference may always result when mating parts are assembled.

Inspection of dimensional and geometrical deviations:-

GD&T tells manufacturing partners and inspectors the allowable variation within the product assembly and standardizes how that variation is measured.

- **Inspection of Dimensional Deviations:**

Calibration and Measurement Tools:

Use calibrated measurement tools such as calipers, micrometers, and gauges to measure linear dimensions accurately.

Gauging:

Implement go/no-go gauges for critical dimensions. These gauges quickly determine whether a part is within acceptable size limits.

Coordinate Measuring Machines (CMM):

CMMs are advanced tools that can measure the coordinates of points on a part's surface, allowing for precise inspection of complex geometries.

Statistical Process Control (SPC): Apply statistical methods to monitor and control the manufacturing process, ensuring that dimensions are within specified tolerances.

Optical Comparators: Optical comparators use optical systems to project an enlarged view of a part onto a screen, facilitating dimensional measurements and comparisons.

- **Inspection of Geometrical Deviations:**

Profile Measurement:

Use profile projectors or profile measurement systems to inspect the contour or shape of a part, ensuring it conforms to the specified profile tolerances.

Coordinate Measuring Machines (CMM) for GD&T:

CMMs are versatile tools that can measure not only linear dimensions but also complex geometrical features, allowing for comprehensive GD&T inspection.

Form Tracers:

Form tracers are devices that follow the contour of a part and provide information on deviations from the specified form.

Surface Roughness Measurement:

For features with specified surface finish requirements, use surface roughness measurement tools to ensure compliance.

Inspection of Angular Deviations:

Employ angle gauges or optical devices to measure angular features and verify that they meet the specified angular tolerances.

Runout Measurement:

Runout, whether radial or axial, can be measured using dial indicators or CMMs. It ensures that the rotating part does not wobble excessively.

- **Datum Inspection:**

Datum Feature Inspection:

Inspect datum features using calibrated measurement tools to ensure their correct size and form.

Datum Reference Frame Verification:

Verify the establishment of the datum reference frame (DRF) and the correct usage of datums in accordance with GD&T principles.

- **Reporting and Documentation:**

Dimensional Inspection Reports:

Document all measured dimensions and deviations in an inspection report for traceability and future reference.

GD&T Inspection Reports:

Include GD&T inspection results, highlighting deviations from specified tolerances and providing information on the form, orientation, and location of features.

Non-Conformance Reporting:

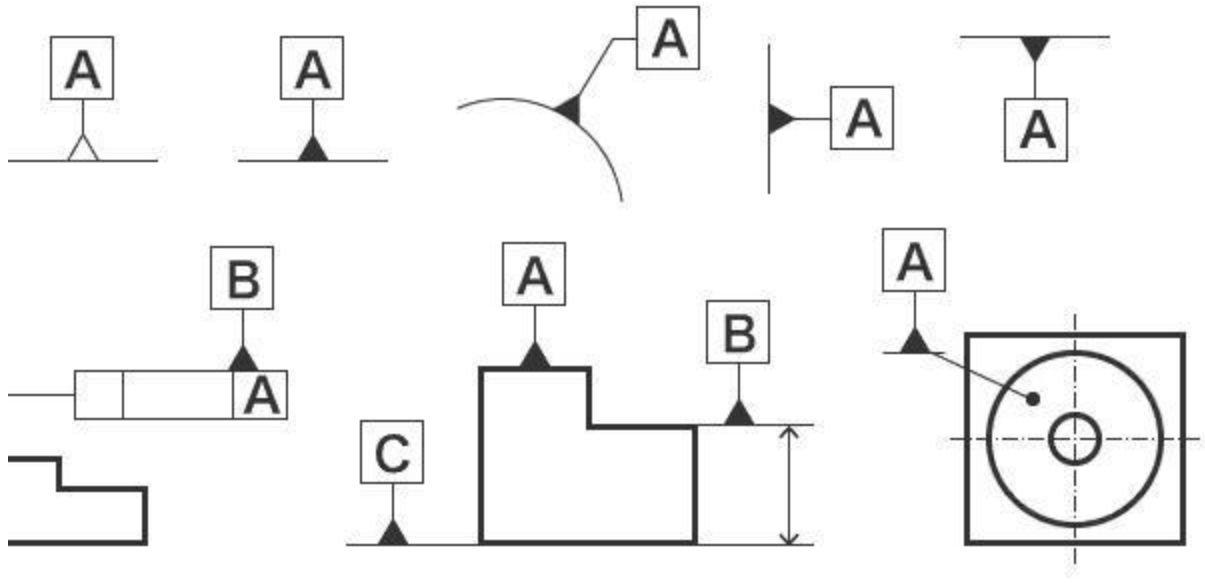
If any part is found to be out of tolerance, initiate non-conformance reporting procedures to address and rectify the issue.

Datums:-

In Geometric Dimensioning and Tolerancing (GD&T), the datum system establishes reference planes and points to define the geometric relationship of features on a part. It consists of primary, secondary, and tertiary datums, aiding in the specification of tolerances for size, form, orientation, and location of features. Primary datums are foundational references, while secondary and tertiary datums refine the alignment. The datum feature symbol is used to identify these references on engineering drawings.

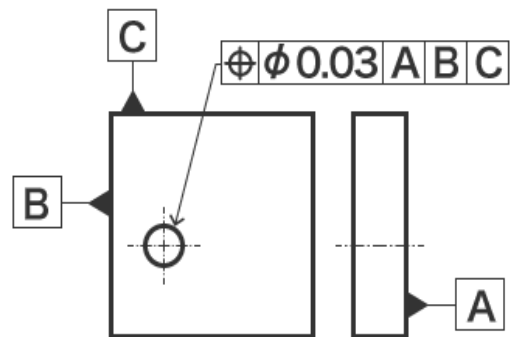
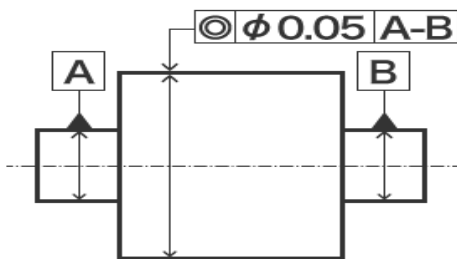
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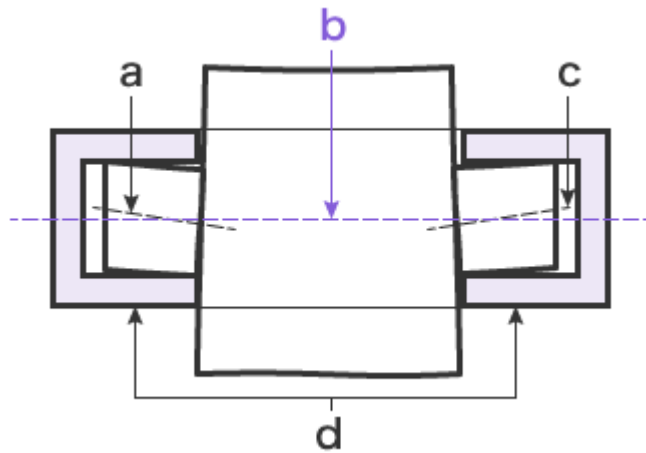
A theoretically exact point, axis, or plane derived from the true geometric counterpart of a specified datum feature



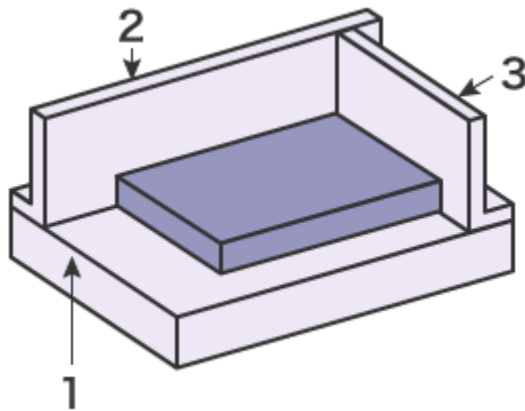
Datums systems:-

A datum system is a group of datums created by combining more than one individual datum in order to use it as a reference for tolerance features. For example, three datums are used to completely capture the orientation of a part.





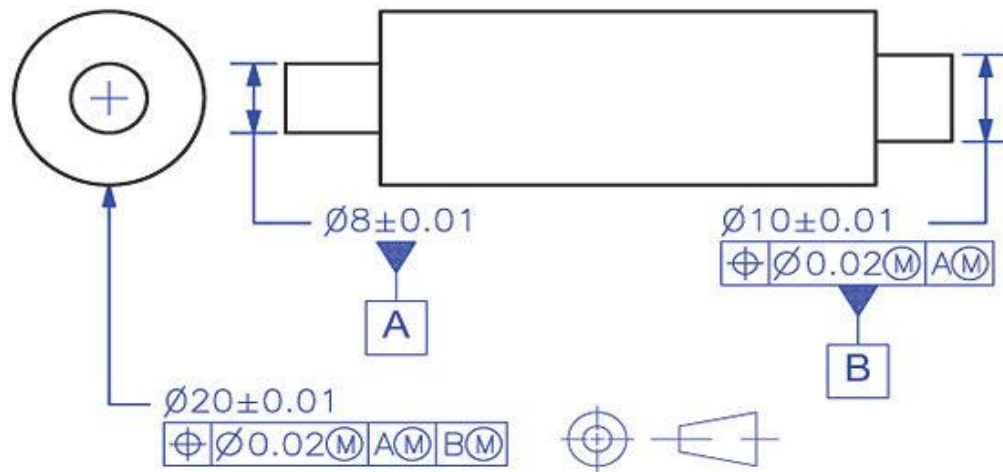
- a. Axis of datum A
- b. Common datum A-B
- c. Axis of datum B
- d. Practical datum feature (i.e., contact surface)



- 1. Push datum A against target feature
- 2. Push datum B against target feature
- 3. Push datum C against target feature

selections of datums:-

Datums should be selected in order of importance, depending on characteristics such as mating surfaces, easy-to-see surfaces, functional part surfaces, and surfaces with a large enough surface area to allow for measurements during inspection. Datum features must be distinguishably different features of a part.



The Datum Feature Selection Process: Because Datum Features serve to constrain degrees of freedom and establish coordinate systems relative to which the remaining features of a part are to be oriented and located, they ought in general (but with some exceptions) to be those features which actually constrain it relative to the mating part during an assembly process. Namely at the end of that process, the objective is for the coordinate systems of two mating parts to coincide in order to guarantee functional relationships between their remaining features. In the examples that follow, this will be the first order of business.

The selection of datums in GD&T is a critical aspect of defining the reference framework for the measurement and tolerancing of features on a part. When choosing datums, consider the functional and manufacturing requirements of the part. Common practices include selecting datums based on:

1. ***Functional Significance:** Choose datums that are relevant to the functional requirements of the part. This ensures that the critical features are properly aligned and controlled.

2. ***Accessibility and Stability:** Select datums that are easily accessible and stable during the measurement process. This helps in achieving consistent and repeatable measurements.

3. *Manufacturability:* Consider the ease of manufacturing and inspecting the selected datums. Datums that align with primary machining or assembly surfaces are often preferred.

4. *Datum Hierarchy:* Establish a hierarchy among the selected datums. Primary, secondary, and tertiary datums define the order in which the part is located and constrained.

5. *Datum Simplicity:* Aim for simplicity in datum selection. Minimize the number of datums to avoid unnecessary complexity in the inspection and manufacturing processes.

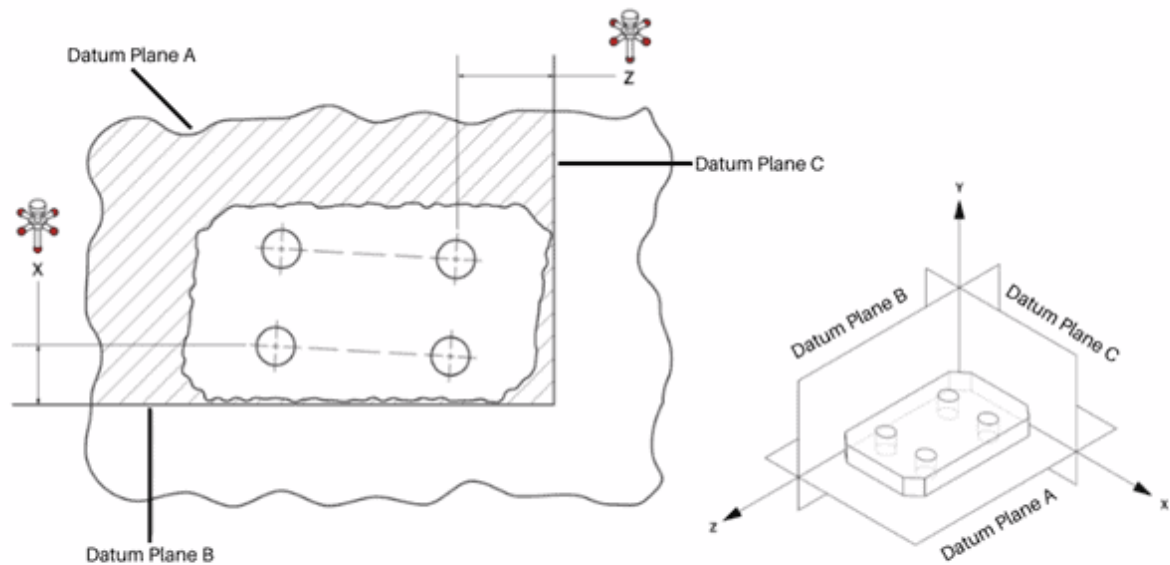
Restraining degrees of freedom:-

In the context of engineering and design, restraining degrees of freedom refers to limiting or controlling the movement or rotation of a part or object along certain axes. This is crucial for ensuring that a component fits and functions correctly within an assembly. The concept is often applied in fields like mechanical engineering, where precise alignment and control of movement are essential.

For example, in a three-dimensional space:

1. ***Translation Degrees of Freedom:*** Restraining movement along the X, Y, and Z axes.
2. ***Rotational Degrees of Freedom:*** Restraining rotation around the X, Y, and Z axes.

By strategically selecting and constraining degrees of freedom, engineers ensure that a part or assembly behaves as intended without unwanted motion or misalignment. This is fundamental to achieving the desired functionality, structural integrity, and overall performance of a system. Techniques such as fixtures, constraints, and geometric features in CAD (Computer-Aided Design) are employed to effectively restrain degrees of freedom during the design and manufacturing processes.



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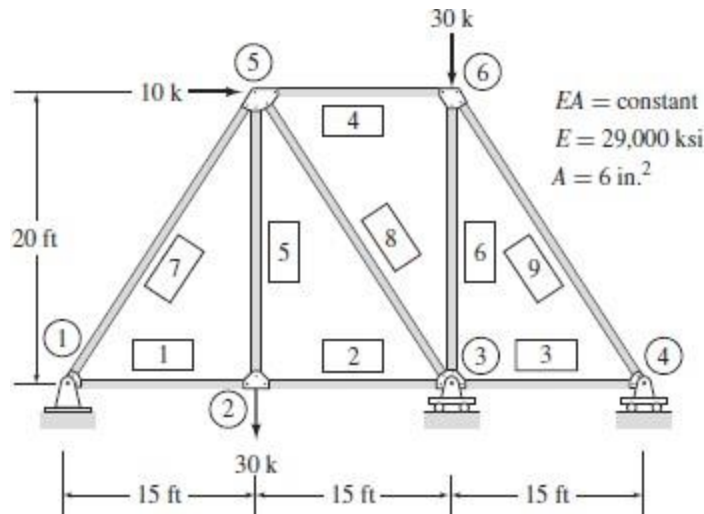
DOF (Degrees of freedom):-

In Geometric Dimensioning and Tolerancing (GD&T), degrees of freedom (DOF) refer to the number of independent movements or rotations a part or feature can have. When applying GD&T, the goal is to control these degrees of freedom to ensure proper functionality, assembly, and interchangeability.

For example:

1. *Translation DOF:* Controlling movement along the X, Y, and Z axes.
2. *Rotation DOF:* Controlling rotation around the X, Y, and Z axes.

By specifying geometric controls such as flatness, straightness, circularity, and parallelism, GD&T helps constrain these degrees of freedom, guiding the manufacturing and inspection processes. This ensures that parts fit together correctly, meet functional requirements, and maintain the necessary relationships between features. The selection and application of datums in GD&T play a crucial role in establishing the reference framework for controlling degrees of freedom.



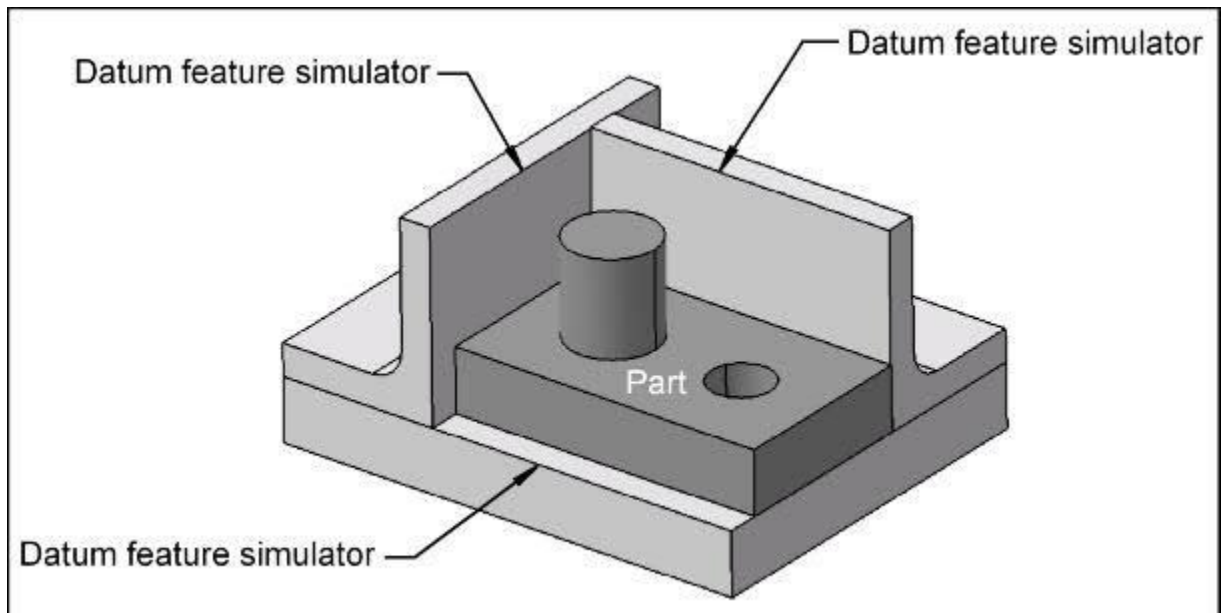
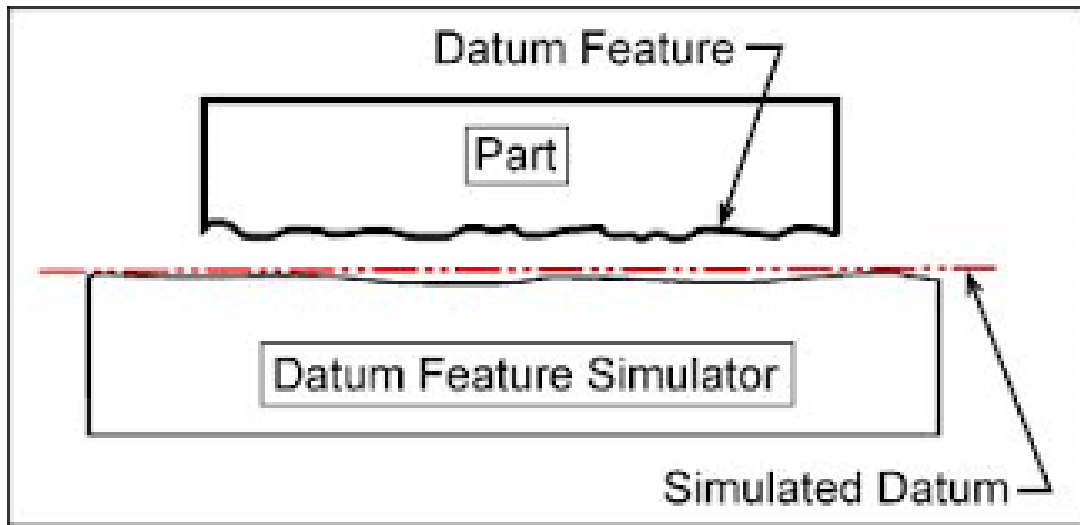
Simulators:-

A datum simulator is a surface of adequate precision oriented to the high points of a designated datum from which the simulated datum is established. Examples: gage pin, block, surface of granite block.

In the context of Geometric Dimensioning and Tolerancing (GD&T), a simulator is a tool or software that allows users to virtually simulate and evaluate the impact of geometric tolerances on a part or assembly. GD&T simulators help users understand how variations in dimensions and tolerances may affect the functionality, fit, and assembly of mechanical components.

These simulators typically provide a visual representation of the part or assembly, allowing users to apply GD&T principles and observe the resulting changes in dimensions and tolerances. Users can simulate different scenarios, such as applying tolerances to specific features or adjusting datum references, and see the effects in real-time.

GD&T simulators can be valuable for training purposes, helping individuals learn and practice GD&T concepts in a virtual environment before applying them to actual design and manufacturing processes. They contribute to better comprehension of how geometric tolerances influence the behavior of parts in practical applications.



Rule #1 (Boundary principle):-

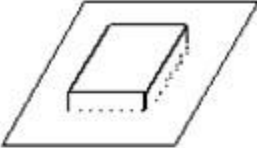
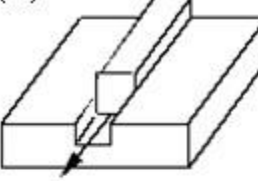
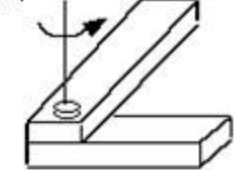
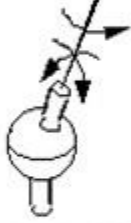
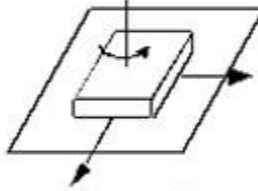
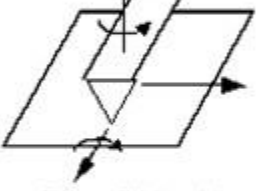
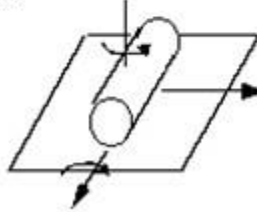
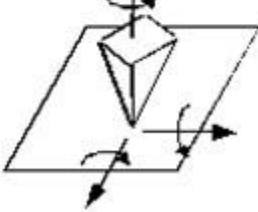
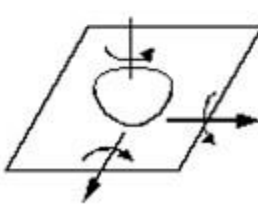
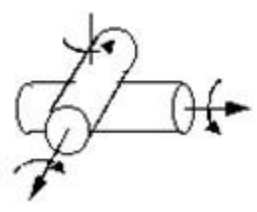
In Geometric Dimensioning and Tolerancing (GD&T), the Boundary Rule or Rule #1 is a fundamental principle that states: "A tolerance specified on a feature of size establishes a boundary that shall be the extreme limit of size for that feature."

In simpler terms, Rule #1 defines that when a tolerance is applied to a feature of size (like a hole diameter or a shaft diameter), the stated tolerance values represent the acceptable range o

sizes for that feature. The actual dimension of the feature must fall within this specified tolerance zone to be considered acceptable.

For example, if a hole is dimensioned as $\text{Ø}25.0 +0.2/-0.1$, the lower limit is 24.9 ($25.0 - 0.1$), and the upper limit is 25.2 ($25.0 + 0.2$). Any hole size within this range is acceptable according to Rule #1.

Understanding and applying Rule #1 is crucial for ensuring that parts and assemblies meet the required specifications and function as intended within the specified tolerance limits.

(a)  Rigid (no motion)	(b)  Prismatic (1)	(c)  Revolute (1)	(d)  Parallel Cylinders (2)
(e)  Cylindrical (2)	(f)  Spherical (3)	(g)  Planar (3)	(h)  Edge Slider (4)
(i)  Cylindrical Slider (4)	(j)  Point Slider (5)	(k)  Spherical Slider (5)	(l)  Crossed Cylinders (5)

Rule #2:-

In Geometric Dimensioning and Tolerancing (GD&T), Rule #2 of the Boundary Principles is related to the application of geometric tolerances. These principles provide guidelines for understanding and applying GD&T effectively. Rule #2 specifically addresses the concept of "Enveloping Principle." Here's a brief explanation:

Rule #2 - Enveloping Principle:

- The envelope of perfect form or boundary shall be uniformly disposed about the true geometric form, with no intentional offset.

In simpler terms, this rule emphasizes that the tolerance zone, which defines the acceptable variation in the dimensions or form of a feature, should uniformly surround the true geometric form without intentional offset. It ensures that the tolerance zone encompasses the entire allowable variation around the perfect form without favoring one side over the other.

Adhering to Rule #2 helps in accurately representing the intended geometry and maintaining consistent control over the part's features within the specified tolerance limits. This principle is essential for effective communication between design, manufacturing, and inspection teams in ensuring the proper fit and function of components.

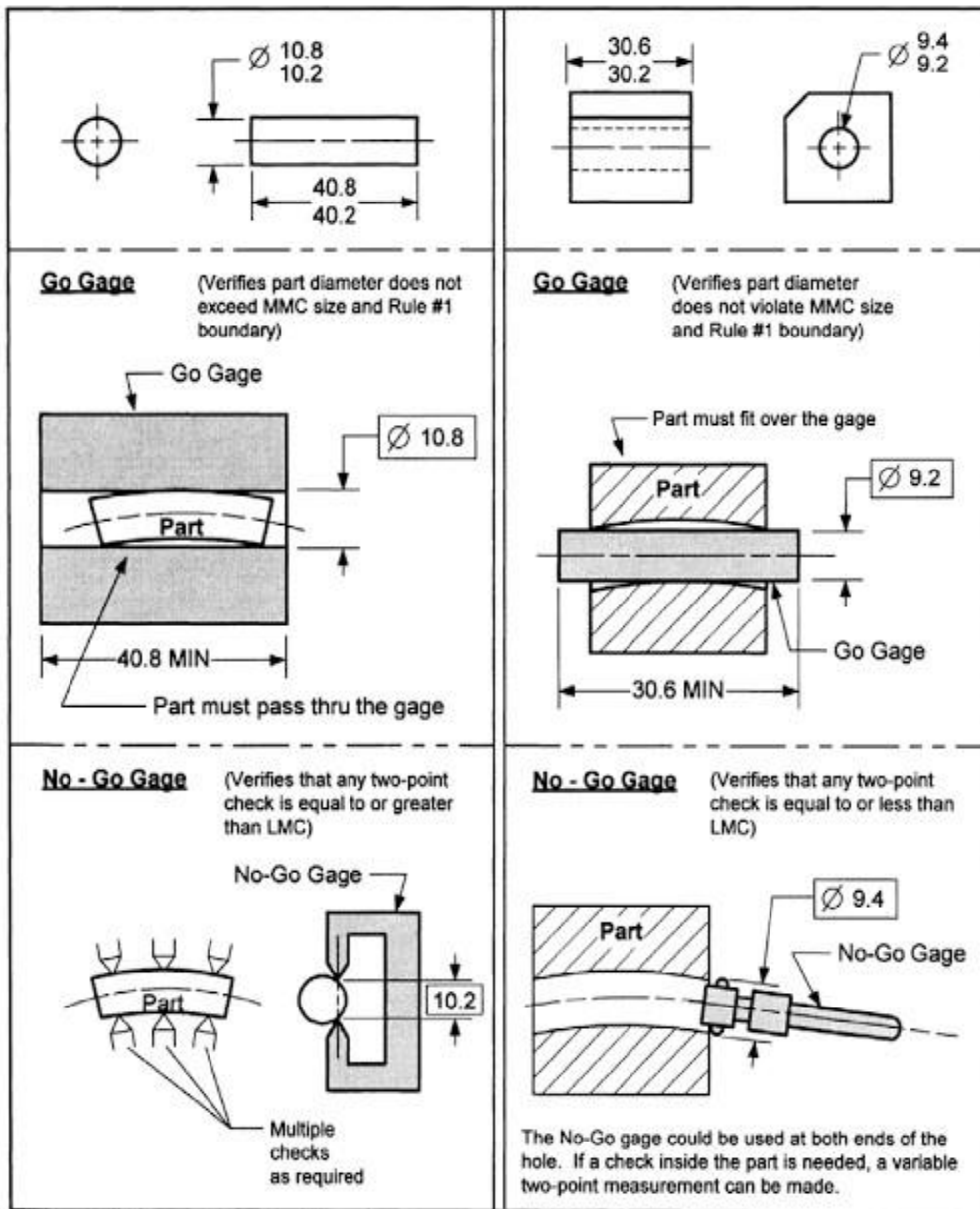


FIGURE 3-4 Go Gage and No-Go Gage Examples